

Unit 1: Computer Networks and Transport Layer Protocols

1. Transmission Control Protocol (TCP)

Transmission Control Protocol (TCP) is a connection-oriented transport layer protocol that provides reliable, ordered, and error-checked delivery of a stream of octets between applications running on hosts communicating via an IP network. TCP uses a three-way handshake process to establish a connection before data transmission begins. The three-way handshake consists of SYN, SYN-ACK, and ACK packets. TCP incorporates flow control using a sliding window protocol to ensure that a sender does not overwhelm a receiver. Furthermore, TCP employs congestion control algorithms such as Slow Start, Congestion Avoidance, Fast Retransmit, and Fast Recovery to prevent network congestion collapse.

2. User Datagram Protocol (UDP)

User Datagram Protocol (UDP) is a connectionless transport layer protocol that emphasizes low latency over reliability. Unlike TCP, UDP does not guarantee packet delivery, ordering, or duplicate protection. UDP headers are lightweight with a fixed size of 8 bytes, consisting of Source Port, Destination Port, Length, and Checksum. UDP is widely utilized in real-time applications such as video streaming, online gaming, DNS queries, and VoIP where minimal latency is critical.

3. Comparison between TCP and UDP

TCP differs from UDP in that TCP guarantees delivery through acknowledgments and retransmissions, whereas UDP provides best-effort delivery. TCP has a larger header overhead of 20 to 60 bytes compared to UDP's 8 bytes. TCP maintains stateful connection tracking, while UDP remains stateless throughout transmission.

4. Domain Name System (DNS) Resolution

The Domain Name System (DNS) is defined as a hierarchical and decentralized naming system for computers, services, and resources connected to the Internet. DNS translates human-friendly domain names like example.com into numerical IP addresses such as 192.0.2.1. DNS resolution primarily operates over UDP port 53 for standard queries to achieve rapid lookup times, but switches to TCP port 53 when response sizes exceed 512 bytes.

5. Network Topologies and Architectures

A network topology is the schematic description of the arrangement of a network, including its nodes and connecting lines. In a Mesh Topology, every device has a dedicated point-to-point link to every other device, requiring $n(n-1)/2$ physical duplex links. In a Star Topology, each device has a dedicated

point-to-point link only to a central controller, usually called a hub. In a Bus Topology, one long cable acts as a backbone to link all devices via drop lines and taps. In a Ring Topology, each device has a dedicated point-to-point connection with only the two devices on either side of it.